JULIAN VAZQUEZ

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EDUCATION

University of California, San Diego — B.S. Applied Mathematics Fall 2026 – Exp. June 2028
La Jolla, CA

Glendale Community College — A.S. Mathematics Completed 2026
Glendale, CA

Mathematics Coursework: Survey of Abstract Algebra, Linear Algebra, Honors Differential Equations, Multivariable/Vector Calculus, Introduction to Statistics
In Progress (Fall 2026): Introduction to Cryptography (MATH 187A), Mathematical Reasoning / Proofs (MATH 109), Introduction to Probability (MATH 180A)
Planned: Modern Algebra I–II, Discrete Mathematics & Graph Theory, Number Theory I–II, Mathematics of Modern Cryptography

CRYPTOGRAPHY PROJECTS

AES-128 Block Cipher — Implementation from NIST FIPS 197 (C) *In Progress, 2026*

- Implementing AES-128 encryption and decryption from NIST FIPS 197, coding the S-box, ShiftRows, MixColumns arithmetic over GF(2^8), and key expansion by hand with no external cryptographic libraries.
- S-box computed from field inverses and verified against FIPS 197 Table 4. Rewrote the S-box and GF(2^8) multiply to run in constant time: masks replace the branch, and the inverse is computed as x^{254} instead of a table lookup. Verified against the reference on every input and in the compiled assembly.

Cryptopals Cryptographic Challenges — C++ *In Progress, 2026*

- Implementing the cryptanalytic attacks by hand, including single-byte and repeating-key XOR key recovery with character-frequency scoring and Hamming-distance keysize detection, plus AES-ECB mode detection and decryption.
- Working toward Set 8, the abstract-algebra set, which covers attacks on RSA, Diffie-Hellman, and elliptic-curve cryptosystems.

Supply Chain Data Analysis — Python 2026

- Built a pandas/NumPy pipeline over a 180,000-row dataset, preserving nulls rather than imputing them and typing identifiers as text so downstream aggregates stayed accurate.

TECHNICAL SKILLS

Mathematics: Abstract algebra, finite fields, linear algebra, differential equations, probability and statistics
Programming: C, C++, Python (pandas, NumPy, Matplotlib, Seaborn, Plotly), SQL
Tools: Git/GitHub, GCC toolchain and sanitizers, Excel/Google Sheets, Tableau

WORK EXPERIENCE

Mathnasium, Burbank, CA — Mathematics Instructor March 2025 – September 2026

- Taught 20–30 K–12 students per week in geometry, statistics, number sense, and calculus.
- Explained abstract concepts by connecting new material to results students already understood, diagnosing the gap behind a recurring error rather than re-teaching the surface mistake.

LEADERSHIP

Society of Hispanic Professional Engineers (SHPE) — Co-Founder and Treasurer February 2025 – June 2026

- Co-founded the Glendale Community College chapter as one of three founding members, building it from nothing to an active organization.
- Secured state funding through written proposals. Maintained chapter balance sheets and managed budgets for chapter events.

CERTIFICATIONS

CompTIA Security+ (SY0-701) — In Progress, October 2026
Google Data Analytics Professional Certificate — Completed August 2026